ES 1C03 · ENVIRONMENTAL SCIENCE

Fast Fashion: *Environmental Impact* in Canada

THE HIDDEN COST OF CHEAP CLOTHING

Fast fashion refers to the rapid production of inexpensive clothing that mimics current luxury fashion trends, designed for short-term use and quick disposal.[1] Canada is one of the highest per-capita consumers of textiles in the world. Canadians purchase an estimated **70 new garments per person annually**, yet the average garment is worn only **7–10 times** before being discarded.[2,3] The environmental consequences of this linear "take-make-dispose" model are severe, contributing to **greenhouse gas emissions, water pollution, microplastic contamination, and massive textile waste** in Canadian landfills.[4] The fashion industry alone accounts for approximately **8–10% of global carbon emissions** — more than international flights and maritime shipping combined.[5]

01 The Fast Fashion Lifecycle

RAW MATERIALS ▸ MANUFACTURING ▸ TRANSPORT ▸ CONSUMER USE ▸ DISPOSAL

02 Key Statistics

12 kg
Textile waste per Canadian per year
(ECCC, 2023)[1]

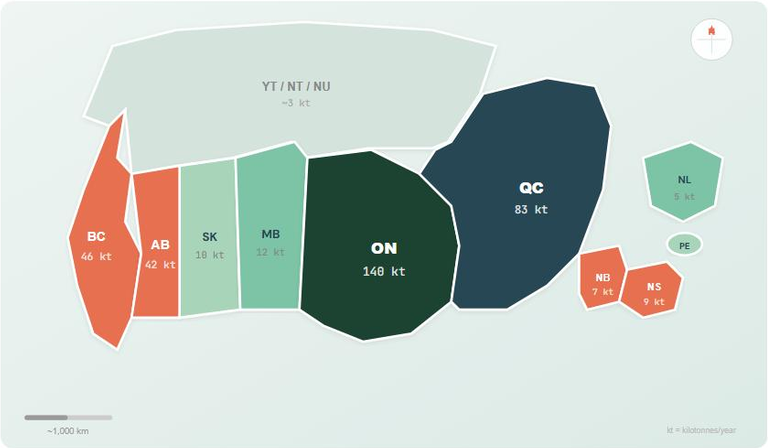
2,700 L
Water to produce one cotton T-shirt
(EMF, 2017)[3]

1.2 Gt
CO_2 from global fashion industry/yr
(UNEP, 2023)[4]

<1%
Clothing recycled into new garments
(EMF, 2017)[3]

03 Data Visualizations

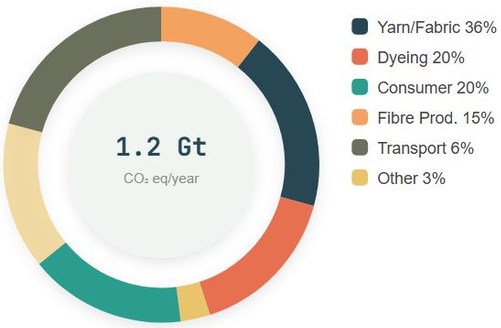
FIGURE 1 · TEXTILE WASTE BY PROVINCE (KILOTONNES/YR)



>100 kt · 30–100 kt · 10–30 kt · <10 kt

Figure 1. Estimated annual textile waste by province, population-weighted. Ontario and Quebec generate the most due to higher populations.[1,2]

FIGURE 2 · FASHION GHG EMISSIONS BY STAGE



One pair of jeans uses ~7,500 L of water to produce[3]

Figure 2. Data: IPCC (2022); Ellen MacArthur Foundation (2017)[3,5]

04 Environmental Impacts in Canada

Carbon Emissions

Canada's textile sector contributes to GHG emissions across the supply chain. Globally, fashion produces ~1.2 Gt CO_2/yr — more than aviation and shipping combined. Synthetic fibres derived from fossil fuels generate emissions during both production and incineration.[4,5]

Water Pollution

Fashion consumes ~93 billion m³ of water per year globally. Textile dyeing is the 2nd-largest water polluter worldwide. In Canada, wastewater from textile processes introduces heavy metals and toxic chemicals into freshwater ecosystems.[3,4]

Landfill & Waste

Canadians send ~12 kg of textile waste per person to landfills annually — over 500,000 tonnes nationally. Synthetics take 20–200 years to decompose, releasing methane. Only ~15% of textiles are diverted from landfills.[1,2]

Biodiversity Loss

Cotton uses 6% of global pesticides and 16% of insecticides. Intensive monoculture causes soil degradation and biodiversity loss. Expansion of fibre crops drives deforestation in regions supplying Canadian retailers.[4,5]

ACTION Solutions: How You Can Help

BUY LESS, CHOOSE WELL
Invest in durable garments. Reducing purchases by 30% could cut emissions equal to removing 1M cars from roads.[3]

THRIFT & SWAP
Shop second-hand or organize swaps. Canada's resale market extends garment lifespans and reduces new production demand.[2]

RECYCLE & DONATE
Use textile recycling programs. Many Canadian cities offer curbside textile collection to divert clothing from landfills.[1]

WASH MINDFULLY
Cold water, full loads, and microfibre-catching bags can reduce microplastic release by up to 80% per cycle.[3]

ADVOCATE FOR POLICY
Support extended producer responsibility (EPR). Canada is developing federal textile waste strategy under Zero Plastic Waste.[1]

SUSTAINABLE FABRICS
Choose organic cotton, TENCEL™, or recycled fibres. Look for GOTS, OEKO-TEX, and Fair Trade certifications.[6]

REFERENCES

[1] Environment and Climate Change Canada (2023) *Reducing plastic and textile waste.* Government of Canada. Available at: https://tinyurl.com/eccc-textile (Accessed: 20 February 2026).
[2] Statistics Canada (2023) *Household spending on clothing and textiles.* Statistics Canada. Available at: https://tinyurl.com/statcan-clothing (Accessed: 20 February 2026).
[3] Ellen MacArthur Foundation (2017) *A new textiles economy: Redesigning fashion's future.* Ellen MacArthur Foundation. Available at: https://tinyurl.com/emf-textiles (Accessed: 20 February 2026).
[4] United Nations Environment Programme (2023) *Sustainability and circularity in the textile value chain.* UNEP. Available at: https://tinyurl.com/unep-textiles (Accessed: 20 February 2026).
[5] Intergovernmental Panel on Climate Change (2022) *Climate Change 2022: Mitigation of Climate Change.* Cambridge University Press. Available at: https://tinyurl.com/ipcc-ar6-wg3 (Accessed: 20 February 2026).
[6] Plastic Collective (2024) *How fast fashion is bad for the environment.* Plastic Collective. Available at: https://www.plasticcollective.co/how-fast-fashion-is-bad-for-the-environment/ (Accessed: 25 February 2026).